

- 6 (a) One of the conclusions from the α -particle scattering experiment was that the atom is mostly empty space.

State the observation from the experiment that provides evidence for this conclusion.

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..... [1]

- (b) An α -particle travelling with a speed of $5.60 \times 10^6 \text{ m s}^{-1}$ collides with a stationary gold nucleus of mass 197 u . After the collision, the α -particle is travelling with a speed of $5.38 \times 10^6 \text{ m s}^{-1}$ in the opposite direction to its initial motion. No external forces act on this system.

- (i) Show that the momentum of the α -particle before the collision is $3.72 \times 10^{-20} \text{ kg m s}^{-1}$.

[2]

- (ii) Use conservation of momentum to calculate the speed of the gold nucleus after the collision.

speed = m s^{-1} [2]

- (iii) The absolute uncertainty in each of the measured speeds of the α -particle is $\pm 5 \times 10^4 \text{ m s}^{-1}$.

Calculate the percentage uncertainty in the change in velocity of the α -particle.

percentage uncertainty =% [2]

- (c) Determine the number of up quarks in a nucleus of ${}^2_1\text{H}$.

number = [1]

- (d) Describe the general quark composition of a meson.

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..... [1]

[Total: 9]